

# Introduction to Python:-



## What is python?

Python is a **high-level programming language** celebrated for its simplicity, readability, and versatility. It means it is used to create programs used in a system, software, applications to perform some specific functions.

It is a widely used language because of its easy syntax like the English language. It was designed by Guido van Rossum and released in 1991.

Unlike, other programming language like c++, c# and java .It also has the feature of oops paradigm which supports the four pillars of oops like inheritance, abstraction, encapsulation and polymorphism.

It is also called as cold scripting language or glue language because it easily integrates with html and other sources to provide better functionality and interface.

It generally saves the time of programmers by increasing the readability and often requires easy syntax.

## Why do we use it?

- 1.We use python due to its some extra supported features:-
2. We can create dynamic web applications
3. web development (server-side),
4. software development,
5. mathematics,
6. system scripting.

## **Why is there a need for python?**

python is needed due to many reasons:-

### **1. Data Science**

Python is the preferred programming language of most data scientists. Be it IT ops, software development or marketing, currently every job makes use of data and depends on it to drive their operations. With the release of 'Numpy' and 'Pandas', Python rose to prominence in the world of data.

### **2. Easy to Learn**

Python is an easy language to master. This is chiefly because of its resemblance to the English language. Python's syntax is characterized by very few rules and special cases.

### **3. Cross-Platform and Open Source**

It's been more than 20 years since this language has been running cross-platform and open source. Be it Linux, Windows or MacOS, Python code works on every platform.

### **4. Vast Libraries**

Python is supported by PyPI which has 85,000+ python scripts and modules accessible to the user. These modules provide pre-packaged functionality available to the users in their local Python environment.

### **5. Flexibility**

Python has several powerful applications integrated with other programming languages. Details about these are given as follows:

.Net and C# compatible: IronPython

A version with C: CPython

Python combined with Ruby: RubyPython

Python integrated with Java: Jython

Python written with Objective C toolkits: PyObjc

### **6. Artificial Intelligence**

Artificial Intelligence (AI) is a pivotal domain where Python shines. Python's libraries like TensorFlow and Keras enable the implementation of complex machine learning and deep learning algorithms

### **7. Web Development**

Python has a variety of framework support website or web-development. Python has an array of frameworks for developing websites.

Popular frameworks such as Django, Flask, and Pylons are characterized by faster and stable code

## **8. Computer Graphics**

Python can be employed in small, large, online or offline projects. It is used to develop GUI and desktop applications.

## **9. Scripting and Automation**

What most people don't know about Python is that it can be used as a scripting language. In scripting, the code is written in script form and gets executed.

## **10. Versatile Language and Platform**

Python remains very relevant today as it can be used in any operations scenario or software development, be it in managing local and cloud infrastructure, working against a SQL database, developing a custom function for Hive & Pig, supporting object-oriented design or even developing a small tool for the user.